Universidad de San Carlos de Guatemala
Facultad de Ingeniería · Escuela de Ciencias y Sistemas

Actividad Práctica 5
Enrutamiento Estático

Curso:	Redes de Computadora 1	**Carnet:**	202200137
Actividad:	Act. Práctica 5 – Enrutamiento Estático	**Estudiante:**	Johanna Alexandra Pérez Enriquez

1. Topología de Red

La red consta de tres routers conectados en cadena. El Router 1 administra la LAN izquierda (PC0 y PC1), el Router 0 es el nodo central de tránsito, y el Router 2 administra la LAN derecha (PC2 y PC3). Los enlaces entre routers usan subredes /30.

- Router 1 – LAN 192.168.37.0/24 (PC0, PC1)
- Router 0 – Nodo central de tránsito
- Router 2 – LAN 10.0.37.0/24 (PC2, PC3)

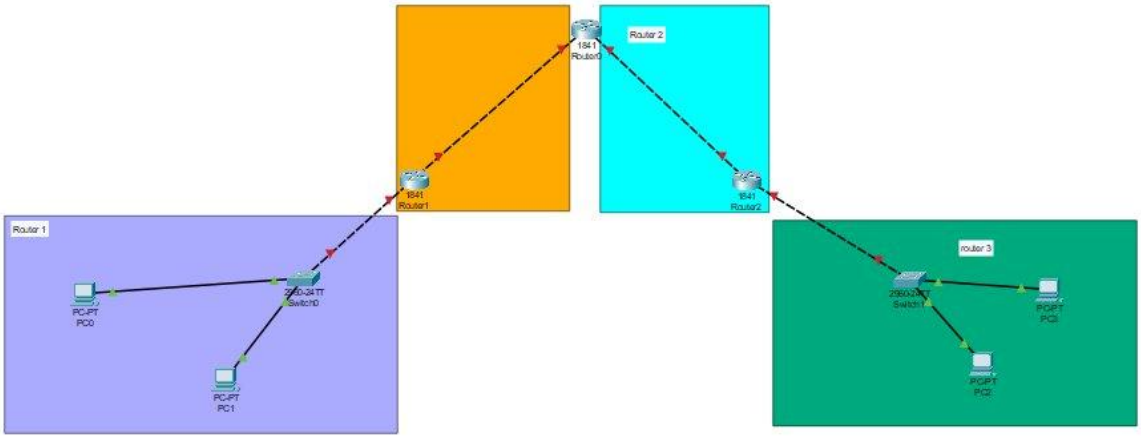

Figura 1 – Topología de red en Cisco Packet Tracer

2. Tabla de Direccionamiento IP

Routers

Dispositivo	Interfaz	Dirección IP	Máscara	Rol
Router 1	Fa0/0	11.0.37.1	255.255.255.252	WAN → R0
Router 1	Fa0/1	192.168.37.1	255.255.255.0	LAN izq.
Router 0	Fa0/0	11.0.37.2	255.255.255.252	WAN → R1
Router 0	Fa0/1	192.167.37.1	255.255.255.252	WAN → R2
Router 2	Fa0/0	192.167.37.2	255.255.255.252	WAN → R0
Router 2	Fa0/1	10.0.37.1	255.255.255.0	LAN der.

PCs

Dispositivo	Dirección IP	Máscara	Gateway
PC0	192.168.37.2	255.255.255.0	192.168.37.1
PC1	192.168.37.3	255.255.255.0	192.168.37.1
PC2	10.0.37.2	255.255.255.0	10.0.37.1
PC3	10.0.37.3	255.255.255.0	10.0.37.1

3. Configuración de Routers

3.1 Router 1

```
enable
configure terminal
hostname Router1

interface FastEthernet0/0
 ip address 11.0.37.1 255.255.255.252
 no shutdown
exit

interface FastEthernet0/1
 ip address 192.168.37.1 255.255.255.0
 no shutdown
exit

ip route 10.0.37.0 255.255.255.0 11.0.37.2
exit
write memory
```

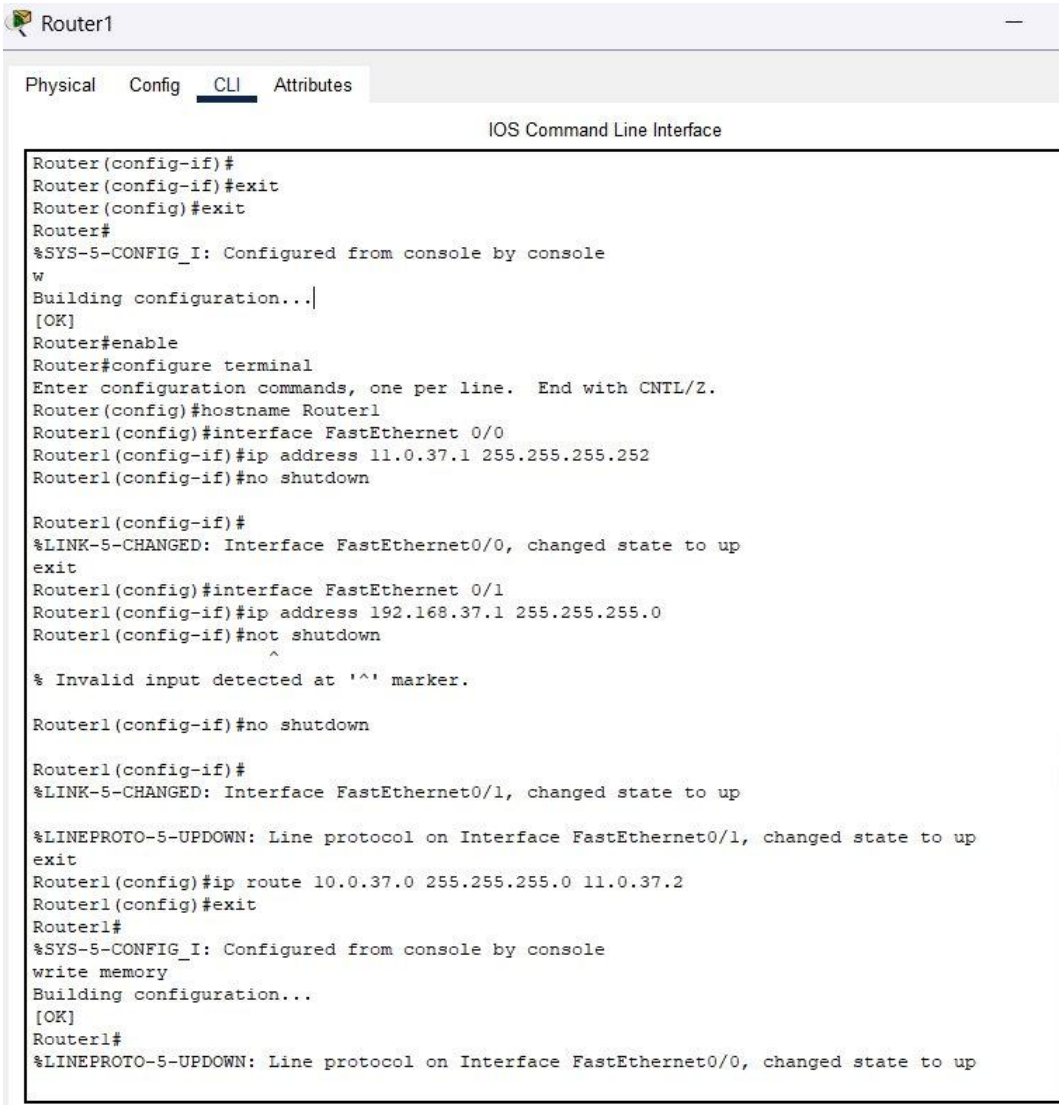

Figura 2 – CLI Router 1: configuración de interfaces y ruta estática

3.2 Router 0 (central)

```
enable
configure terminal
hostname Router0

interface FastEthernet0/0
 ip address 11.0.37.2 255.255.255.252
 no shutdown
exit

interface FastEthernet0/1
 ip address 192.167.37.1 255.255.255.252
 no shutdown
exit

ip route 192.168.37.0 255.255.255.0 11.0.37.1
ip route 10.0.37.0    255.255.255.0 192.167.37.2
exit
write memory
```

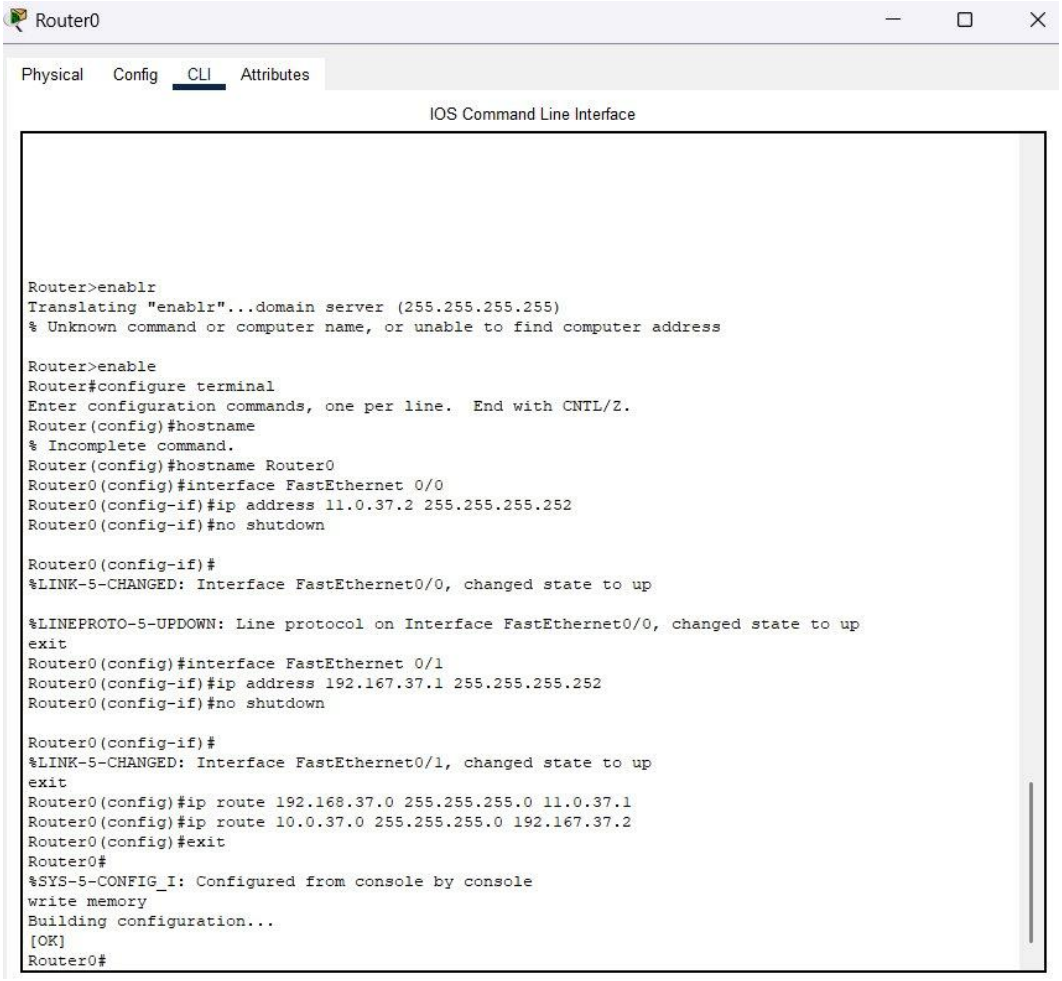

Figura 3 – CLI Router 0: configuración y rutas estáticas

3.3 Router 2

```
enable
configure terminal
hostname Router2

interface FastEthernet0/0
 ip address 192.167.37.2 255.255.255.252
 no shutdown
exit

interface FastEthernet0/1
 ip address 10.0.37.1 255.255.255.0
 no shutdown
exit

ip route 192.168.37.0 255.255.255.0 192.167.37.1
exit
write memory
```

3.4 Comandos de Verificación

```
show running-config       ! configuracion activa
show ip route             ! tabla de enrutamiento
show ip interface brief   ! estado de interfaces
```

4. Pruebas de Conectividad

4.1 Desde PC0 → red derecha (ping + tracert)

PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
C:\>ping 10.0.37.2

Pinging 10.0.37.2 with 32 bytes of data:

Reply from 10.0.37.2: bytes=32 time<1ms TTL=125
Reply from 10.0.37.2: bytes=32 time<1ms TTL=125
Reply from 10.0.37.2: bytes=32 time<1ms TTL=125
Reply from 10.0.37.2: bytes=32 time<1ms TTL=125

Ping statistics for 10.0.37.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>tracert 10.0.37.2

Tracing route to 10.0.37.2 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.37.1
  2   0 ms      0 ms      0 ms      11.0.37.2
  3   0 ms      0 ms      0 ms      192.167.37.2
  4   0 ms      0 ms      0 ms      10.0.37.2

Trace complete.

C:\>tracert 10.0.37.3

Tracing route to 10.0.37.3 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.37.1
  2   0 ms      0 ms      0 ms      11.0.37.2
  3   0 ms      0 ms      0 ms      192.167.37.2
  4   *         1 ms      0 ms      10.0.37.3

Trace complete.

C:\>
```

Figura 4 – PC0: ping a 10.0.37.2 (0 % pérdida) y tracert de 4 saltos

4.2 Desde PC2 → red izquierda (ping + tracert)

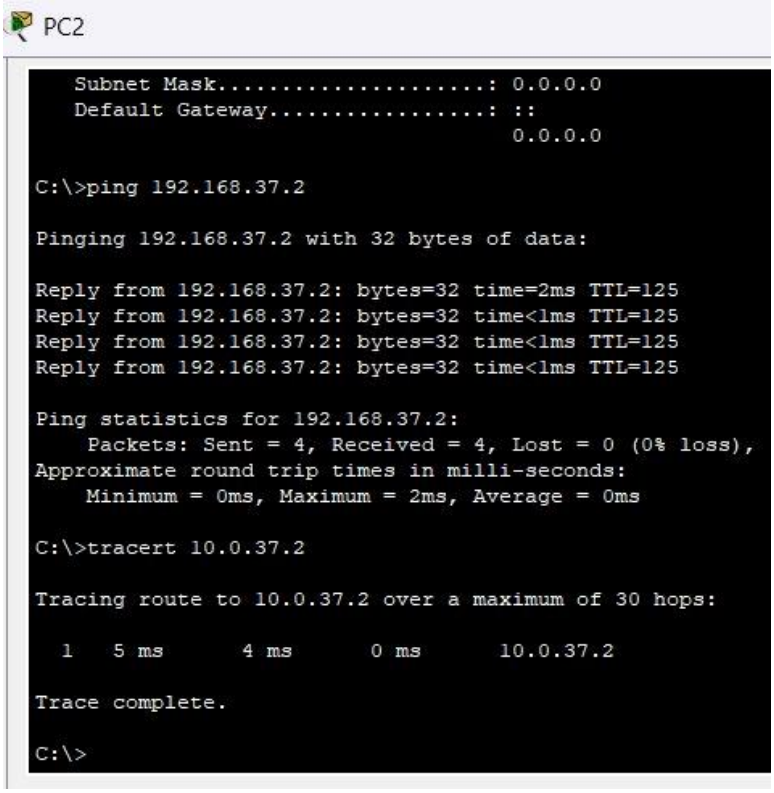

Figura 5 – PC2: ping a 192.168.37.2 (0 % pérdida) y tracert

4.3 Resumen de resultados

Origen	Destino	Prueba	Resultado
PC0 192.168.37.2	PC2 10.0.37.2	ping	OK – 0% perdida
PC0 192.168.37.2	PC3 10.0.37.3	ping	OK – 0% perdida
PC0	10.0.37.2	tracert	4 saltos
PC2 10.0.37.2	PC0 192.168.37.2	ping	OK – 0% perdida
PC2	10.0.37.2	tracert	1 salto (local)